Problem 01: Agentic AI System for Research Paper Discovery, Verification, and Scientific Reasoning

Domain: Generative AI | NLP | Information Retrieval | Agentic AI
Difficulty: Hard

Problem

Researchers manually search hundreds of papers to understand a topic, compare methods, find limitations and spot gaps. Search tools return relevant documents but do not connect evidence across papers or separate established findings from unsupported claims.

Objective

Build an interactive platform that takes a research question and PDFs and produces an evidence-backed report: retrieve, extract findings, compare methods, verify claims, and suggest research gaps.

Users

Graduate students, researchers, literature-review writers.

Core Requirements

1. Accept research questions and PDF uploads.
2. Parse papers into sections: methodology, experiments, results, limitations.
3. Hybrid retrieval: BM25 keyword search plus dense embeddings.
4. Reranker (cross-encoder) to improve passage relevance.
5. Answers with paragraph-level citations.
6. Compare experimental results and methods across papers.
7. Detect contradictions; separate conflicting evidence from consensus.
8. Claim verification against retrieved evidence.
9. Follow-up questions with conversation memory.
10. Uncertainty estimates, or abstain when evidence is insufficient.

Advanced Challenges

- Agentic retrieval: query decomposition and iterative search.
- Graph of papers, methods and findings.
- Citation verification and unsupported-claim detection.
- Evaluate retrieval quality, answer faithfulness and citation correctness.

Suggested Datasets

S2ORC, arXiv, Semantic Scholar API, SciFact / SciFact-Open (claim verification), QASPER (paper QA), PubMedQA.

Suggested Tech

Python, LangGraph, LangChain, ChromaDB or Qdrant, rank_bm25, sentence-transformers / cross-encoders, GROBID or PyMuPDF for parsing, open-source LLMs, Streamlit or React.

Reference Architecture

PDF parser → section chunker → BM25 + vector index → hybrid retriever → reranker → planner agent (decompose) → evidence synthesizer → verifier agent (claim ↔ passage entailment) → report generator → UI.

Evaluation

Recall@k / nDCG (retrieval), RAGAS-style faithfulness, citation precision/recall, claim-verification F1 on SciFact, abstention accuracy. Baselines: BM25 only, dense only, no-reranker, plain LLM without retrieval.

Expected Deliverable

Structured report: literature summary, method/result comparison table, research gaps, citations, limitations and uncertainty analysis.

Pitfalls

Hallucinated citations; chunking that splits tables from captions; only testing on easy questions.

Suggested Timeline (48h)

0-8h parsing + index; 8-20h retrieval + rerank; 20-32h generation + citations; 32-40h verification + contradictions; 40-48h eval + demo.

Standard Judging Rubric (applies to every problem)

Criterion	Weight
Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%

Criterion	Weight
Presentation and demo	5%

Mandatory Submission Checklist

- ☐GitHub repository with setup instructions
- ☐Working prototype or recorded demo
- ☐Documented system architecture
- ☐Dataset sources and preprocessing methodology
- ☐Evaluation results with baseline comparisons
- ☐Limitations and failure cases
- ☐One technically challenging feature demonstrated
- ☐Short presentation: problem, approach, results, future work